



QUERYING DATA FROM A TABLE

SELECT c1, c2 **FROM** t;

Query data in columns c1, c2 from a table

SELECT * FROM t;

Query all rows and columns from a table

SELECT c1, c2 **FROM** t

WHERE condition;

Query data and filter rows with a condition

SELECT DISTINCT c1 **FROM** t

WHERE condition;

Query distinct rows from a table

SELECT c1, c2 **FROM** t

ORDER BY c1 **ASC** [**DESC**];

Sort the result set in ascending or descending order

SELECT c1, **aggregate**(c2)

FROM t

GROUP BY c1;

Group rows using an aggregate function

SELECT c1, **aggregate**(c2)

FROM t

GROUP BY c1

HAVING condition;

Filter groups using HAVING clause

QUERYING FROM MULTIPLE TABLES

SELECT c1, c2

FROM t1

INNER JOIN t2 **ON** condition;

Inner join t1 and t2

SELECT c1, c2

FROM t1

LEFT JOIN t2 **ON** condition;

Left join t1 and t2

MANAGING TABLES

CREATE TABLE t (id **INT** **PRIMARY**
KEY, name **VARCHAR** **NOT NULL**,
price **INT** **DEFAULT** 0

);

Create a new table with three columns

USING SQL CONSTRAINTS

CREATE TABLE t(c1 **INT**, c2 **INT**,
c3 **VARCHAR**,
PRIMARY KEY (c1,c2)

);

Set c1 and c2 as a primary key

CREATE TABLE t1(c1 **INT** **PRIMARY KEY**,
c2 **INT**, **FOREIGN KEY** (c2)
REFERENCES t2(c2)

);

Set c2 column as a foreign key

CREATE TABLE t(
c1 **INT**, c1 **INT**,
UNIQUE(c2,c3)

);

Make the values in c1 and c2 unique

CREATE TABLE t(c1 **INT**
PRIMARY KEY,
c2 **VARCHAR** **NOT NULL**

);

Set values in c2 column not NULL



MODIFYING DATA

INSERT INTO t(column_list)
VALUES(value_list);

Insert one row into a table

INSERT INTO t(column_list) **VALUES**
(value_list), (value_list),;

Insert multiple rows into a table

INSERT INTO t1(column_list) **SELECT**
column_list
FROM t2;

Insert rows from t2 into t1

UPDATE t
SET c1 = new_value;

Update new value in the column c1 for all rows

UPDATE t
SET c1 = new_value, c2
= new_value
WHERE condition;

Update values in the column c1, c2 that match
the condition

SQL AGGREGATE FUNCTIONS

AVG returns the average of a list

COUNT returns the number of elements of a
list

SUM returns the total of a list

MAX returns the maximum value in a list

MIN returns the minimum value in a list

General Format SQL-query

```
SELECT <list of attributes>
FROM <table>
WHERE <condition>
GROUP BY <list of attributes>
HAVING <condition>
ORDER BY <list of attributes>;
```

Order of execution

SELECT	5
FROM	1
WHERE	2
GROUP BY	3
HAVING	4
ORDER BY	6